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Hydroxyl radical formation during oxygen-mediated oxidation of ferrous iron on mineral surface: Dependence on mineral identity

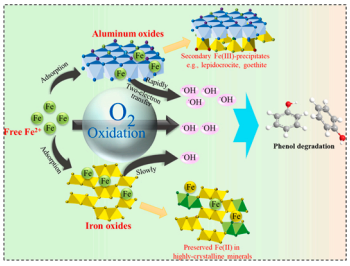
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HIGHLIGHTS

- •OH production was enhanced during Fe(II) oxidation by O2 with addition of Al_2O_3.
- •OH production decreased in Fe(II)/Fe (III)-oxide systems compared to Fe(II) alone.
- Oxidation of surface-sorbed Fe(II) was responsible for the enhanced •OH production.
- Formation of highly-crystalline Fe phases occurred in the Fe(II)/γ-Al2O3 system.
- Mineral properties and pH affected •OH formation and phenol degradation.

GRAPHICAL ABSTRACT



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ABSTRACT

Many studies have examined the redox behavior of ferrous ions (Fe(II)) sorbed to mineral surfaces. However, the associated hydroxyl radical ($^{\bullet}OH$) formation during Fe(II) oxidation by O_2 was rarely investigated at circum-neutral pH. Therefore, we examined $^{\bullet}OH$ formation during oxygenation of adsorbed Fe(II) ($Fe(II)_{sorbed}$) on common minerals. Results showed that 16.7 ± 0.4–25.6 ± 0.3 μM of $^{\bullet}OH$ was produced in Fe(II) and α/γ-Al_2O_3 systems after oxidation of 24 h, much more than in systems with dissolved Fe(II) (Fe^{2+}_{aq}) alone (10.3 ± 0.1 μM). However, $^{\bullet}OH$ production in Fe(II) and α-FeOOH/α-Fe_2O_3 systems (6.9 ± 0.1–8.3 ± 0.1 μM) slightly decreased compared to Fe^{2+}_{aq} only. Further analyses showed that enhanced oxidation of $Fe(II)_{sorbed}$ was responsible for the increased $^{\bullet}OH$ production in the Fe(II)/Al_2O_3 systems. In comparison, less Fe(II) was oxidized in the α-FeOOH/α-Fe_2O_3 systems, which was probably ascribed to the quick electron-transfer between $Fe(II)_{sorbed}$ and Fe(III) lattice due to their semiconductor properties and induced formation of high-crystalline Fe(II) phases that hindered Fe(II) oxidation and $^{\bullet}OH$ formation. The types of minerals and solution pH strongly affected Fe(II) oxidation and $^{\bullet}OH$ production, which consequently impacted phenol degradation. This study highlights that the properties of minerals exert great impacts on surface-Fe(II) oxidation and $^{\bullet}OH$ production during water/soil redox fluctuations.

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1. Introduction

As one of the most powerful reactive oxygen species (ROS), hydroxyl radicals ($^{\bullet}OH$) can unselectively participate in a wide range of redox reactions involving contaminant transformation (Chen et al., 2019; Gligorovski et al., 2015; Huang et al., 2021a; Liu et al., 2017; Zeng et al., 2017), organic matter degradation (Zeng et al., 2020; Zhao et al., 2021), nutrient cycling and greenhouse gases emission (Chen et al., 2021a; Shaw et al., 2021; Zhang et al., 2020). Recent studies have documented that oxidation of ferrous iron (Fe(II)), including the exchangeable, surface-adsorbed or complexed, and mineral-structural Fe(II) species, mainly contributed to $^{\bullet}OH$ formation during soil/sediment redox fluctuations (e.g., surface water and groundwater interactions, water-table decline in paddy soils), and demonstrated that $^{\bullet}OH$ was capable of oxidizing organic matter and contaminants (Chen et al., 2021a, 2021b; Liao et al., 2019; Xie et al., 2020). Among these Fe(II) species, surface-adsorbed Fe(II) ($Fe(II)_{sorbed}$) played a critical role in driving $^{\bullet}OH$ production, with extents affected by $Fe(II)_{sorbed}$ contents, the crystallinity of mineral phases and properties (e.g., pH) of soil/sediment (Chen et al., 2021a, 2021b; Xie et al., 2020). Although such a role of $Fe(II)_{sorbed}$ has been recognized, deeper our understanding of the processes and mechanism of $^{\bullet}OH$ formation involving specific soil minerals is rather limited.

Common minerals, including goethite (e.g., alpha (α)-FeOOH), aluminum oxides (e.g., α-Al_2O_3), and clay minerals (e.g., montmorillonite) are important constituents of soils and highly active to adsorb Fe (II) due to their surface properties (Dzombak and Morel, 1990; Nano and Strathmann, 2006), such as active sites (e.g., hydroxyl groups), acid-base chemistry, electrostatic interactions. Surface-complexed Fe(II) is generally more reductive than aqueous (dissolved) Fe(II) (Fe^{2+}_{aq}), mainly because the donation of electron density from the ligand to Fe(II) enlarges the energy of electron-donating Fe 3*d* orbitals, which enhances the capacity for electron transfer (Huang et al., 2021b). For specific minerals, redox behavior of $Fe(II)_{sorbed}$ was highly impacted by the identity of the structural surfaces (e.g., hematite (001), (110)) and its coordination sites (weak- and strong- binding sites) (Catalano et al., 2010; Nano and Strathmann, 2006), further influencing the recrystallization of precipitates on mineral surfaces. For different minerals, the standard redox potential values (affected by the Gibbs free energy (ΔG^0) values) (Gorski et al., 2016), particle size or aggregation state (Frierdich et al., 2015), and surface defects of associated minerals play an important role in driving oxidation of $Fe(II)_{sorbed}$ (Huang et al., 2021b). Besides, solution conditions (e.g., pH and ionic strength) also exert important influences over the reactivity of $Fe(II)_{sorbed}$ (Huang and Zhang, 2020; Pham and Waite, 2008a). Collectively, these critical factors, associated with mineral properties and solution conditions, predominate the redox behavior of $Fe(II)_{sorbed}$ during geochemical processes, such as electron-transfer to oxidants (e.g., O_2) or organic/inorganic contaminants (e.g., chlorinated alkanes) and stabilization of organic carbon via adsorption/coprecipitation on mineral surfaces (Amonette et al., 2000; Chen et al., 2015; Jones et al., 2014). This knowledge gap becomes even larger when we consider ROS formation and contaminants degradation during Fe(II) oxidation on mineral surfaces.

Although a few studies reported that oxidation of surface-sorbed Fe (II) can produce ROS (e.g., $^{\bullet}OH$) (Jones et al., 2014; Xie et al., 2020), the associated mechanisms and environmental implications have not been systematically investigated yet. Heterogeneous oxidation of surface-Fe (II) has been proposed to follow the Haber-Weiss mechanism (Rose and Waite, 2002), where the $O_2^{\bullet-}$ and H_2O_2 are important intermediates (Eqs. 1–4). Based on aforementioned influencing factors, $^{\bullet}OH$ formation from Fe(II) oxidation can also be strongly affected by solution pH and mineral properties. Compared to oxidation of Fe^{2+}_{aq}, $^{\bullet}OH$ production may be enhanced during oxidation of $Fe(II)_{sorbed}$ due to the elevated electron-transfer via inner-sphere complexation (Rosso and Morgan, 2002). Besides, two-electron transfer pathway might occur during oxidation of surface-Fe(II) (Eq. 5) versus one-electron transfer pathway in oxidation of Fe^{2+}_{aq} (Amonette et al., 2000; Gorski et al., 2016), which is still unsubstantiated yet.

$$Fe(II) + O_2 \rightarrow Fe(III) + O_2^{\bullet-} \quad (1)$$

$$Fe(II) + O_2^{\bullet-} + 2\,H^+ \rightarrow Fe(III) + H_2O_2 \quad (2)$$

$$Fe(II) + H_2O_2 \rightarrow Fe(III) + {}^{\bullet}OH + OH^- \quad (3)$$

$$Fe(II) + {}^{\bullet}OH \rightarrow Fe(III) + OH^- \quad (4)$$

$$2Fe(II)_{sobed} + O_2 + 2\,H^+ \rightarrow 2Fe(III)_{sobed} + H_2O_2 \quad (5)$$

The main purpose was to investigate the processes and mechanism of $^{\bullet}OH$ formation and organic contaminant degradation during oxygenation of $Fe(II)_{sorbed}$ on common minerals (e.g., α-Al_2O_3, gamma (γ)-Al_2O_3, α-FeOOH, α-Fe_2O_3). The $^{\bullet}OH$ production was quantified under experimental conditions, with the underlined mechanism explored based on the analyses of Fe speciation and electron-transfer pathways. State-of-the-art characterization (e.g., X-ray photoelectron spectroscopy (XPS), Mössbauer spectra) was applied to investigate the recrystallization and transformation of Fe phases. To elucidate environmental implications of $^{\bullet}OH$ production, phenol was selected as model compound to investigate its degradation due to its wide distribution in water and soil environments.

2. Materials and methods

2.1. Chemicals and materials

The α-Al_2O_3 (99.9%), γ-Al_2O_3 (99.9%) and α-Fe_2O_3 (99.5%) were obtained from the Aladdin Reagent Co., Ltd. (Shanghai, China), with the specific surface area (SSA) of 39.2, 122.7 and 159.5 $m^2\,g^{-1}$, respectively. The α-FeOOH was synthesized based on the well-established method (Hiemstra et al., 1989), with the SSA of 62.5 $m^2\,g^{-1}$. The XRD pattern of synthesized α-FeOOH was identical to the standard card based on our previous study (Wu et al., 2020). The SEM images of these minerals are shown in Fig. S1. Other chemicals used in this study are listed in Text A1.

2.2. Fe(II) oxidation experiments

Typical Fe(II) oxidation experiments were performed in triplicates under dark conditions at 25 ± 1 °C. In brief, all reagents were prepared and stored in an anoxic glovebox (Braun Co., Germany). Experiments were conducted in 40 mL brown vials containing 20 mM MES (at pH 5.0 and 6.0) or MOPs (pH at 7.0) with pH adjusted using NaOH or HCl. The MES and MOPs were documented to be non-complexing with metal ions (Kandegedara and Rorabacher, 1999). Then, 40 mg minerals were added into each vials for a loading of 2.0 g L^{-1}, which was followed by the addition of 1 mM Fe(II). Thereafter, theses vials were sealed anoxically and moved out for oxygenation via opening caps for 30 s to allow for air exchange. The dissolved O_2 (9.58 mg L^{-1}) was sufficient for oxygenation. These vials were sealed again and shaken at 150 rpm. Control experiments were also performed under the same conditions without minerals. At predetermined time intervals, a set of vials were sacrificed to analyze Fe species.

For $^{\bullet}OH$ quantitation, 10 mM BA was added as chemical probe before the oxygenation to capture $^{\bullet}OH$ via formation of *p*-hydroxybenzoic acid (*p*-HBA) (Mopper and Zhou, 1990). To examine the participation of $O_2^{\bullet-}$ and H_2O_2, nitro blue tetrazolium (NBT, 1 mM) and catalase (CAT, 100 mg L^{-1}) were added as scavengers. The phenol degradation was examined with addition of 4.8 μm phenol instead of BA before oxygenation.

The effects of pre-equilibrium on Fe(II) oxidation and $^{\bullet}OH$ formation were also investigated under the same conditions. After anoxic seal, these vials were placed on a reciprocating shaker (150 rpm) for 48 h to equilibrate the Fe^{2+}_{aq} and minerals, and then the vials were oxidized via

opening caps as above. Results showed that no significant difference was observed at pH 6.0, with slight increase of •OH formation monitored in pre-equilibrium treatments at pH 7.0 (Fig. S2a). Hence, pre-equilibrium negligibly influenced •OH formation, and batch experiments were performed without pre-equilibrium.

2.3. Quantification and characterization of Fe species

To investigate the changes of Fe species, chemical extraction was applied to dissolve the targeted Fe species. In the first treatment, Fe^{2+}_{aq} was separated via filtering the suspensions through 0.22 μm poly(ether sulfone) membrane (ANPEL, Shanghai, China) after centrifugation at 5,000g, and the obtained Fe^{2+}_{aq} sample was added with 0.1 M HCl to avoid any unexpected Fe(II) oxidation. In the second treatment, paralleled samples were extracted by 0.5 M HCl (0.5 M HCl-Fe) for 2 h to dissolve portions of $Fe(II)_{sorbed}$ or Fe(II) in low-crystalline minerals (formed during oxygenation) (Heron et al., 1994; Voelz et al., 2019), also including $Fe_{2+\ aq}$. In the third treatment, paralleled samples were dissolved by 5 M HCl for 24 h to obtain the total Fe (TFe) (Voelz et al., 2019), with recovery of > 99%. The Fe(II) and TFe were measured with a UV-Vis spectrometer (Shimadzu UV-2550) using the phenanthroline method (Tamura et al., 1974). The Fe_{sorbed} or Fe in low-crystalline minerals were obtained by subtraction of Fe^{2+}_{aq} from 0.5 M HCl-Fe. Likewise, Fe phases in higher-crystallinity were calculated based on subtraction of 0.5 M HCl-Fe from 5 M HCl-Fe.

The mineral samples for characterization were separately prepared. At predetermined time intervals, solids were collected after centrifugation at 10,000g (4 °C). Then, samples were lyophilized and preserved in O_2-impermeable Kapton tape before analyses. X-ray photoelectron spectroscopy (XPS) was used to examine variations of Fe status on mineral surface. Mössbauer spectra were collected at 13 K to further identify the changes of Fe phases during oxygenation. Details of these characterizations are provided in Text A2. To examine the Fe(II) speciation in the presence of γ-Al_2O_3 at different pH, surface complexation modeling was performed on MINEQL+ (Environmental Research Software), with the thermodynamic data for calculation listed in Table S2.

2.4. Other analyses

The concentration of *p*-HBA was measured by a Shimadzu 15 C HPLC equipped with a SPD-15 C UV-Vis detector and an Inter Sustain C18 column (Chen et al., 2018). The •OH concentrations were calculated with a conversion factor of 5.87 (Zhou and Mopper, 1990). For phenol analyses, mobile phase was a mixture of acetonitrile and water (45:55, v/v) at 1 mL min^{-1}, with the detection wavelength at 210 nm.

3. Results and discussion

3.1. The •OH production during oxygenation of Fe(II) in the presence of minerals

For Fe^{2+}_{aq} only (Fig. 1a), a lag time for •OH production was observed within the initial 1 h. The •OH accumulation reached a plateau concentration of 8.7 ± 0.3 μM within 10 h. Lagged •OH production can be attributed to the slow Fe(II) oxidation by O_2 (Eq. 1, $k = 0.11\ M^{-1}s^{-1}$) via the out-sphere mechanism (Pham and Waite, 2008b), which has been documented as the rate-limiting step during oxidation of Fe^{2+}_{aq} (Pham and Waite, 2008b; Qian et al., 2019). With further oxidation, more Fe (III) precipitated as low-crystalline mineral phase (e.g., ferrihydrite) (Chen and Thompson, 2018), observed as yellow precipitates during oxygenation (data not shown). Consequently, Fe(II) oxidation and •OH formation might be accelerated by the newly formed Fe phases via surface-catalyzed heterogeneous reactions at the periphery of active sites (Chen and Thompson, 2018; Rosso and Morgan, 2002). With addition of Al_2O_3 minerals, •OH accumulation significantly enhanced

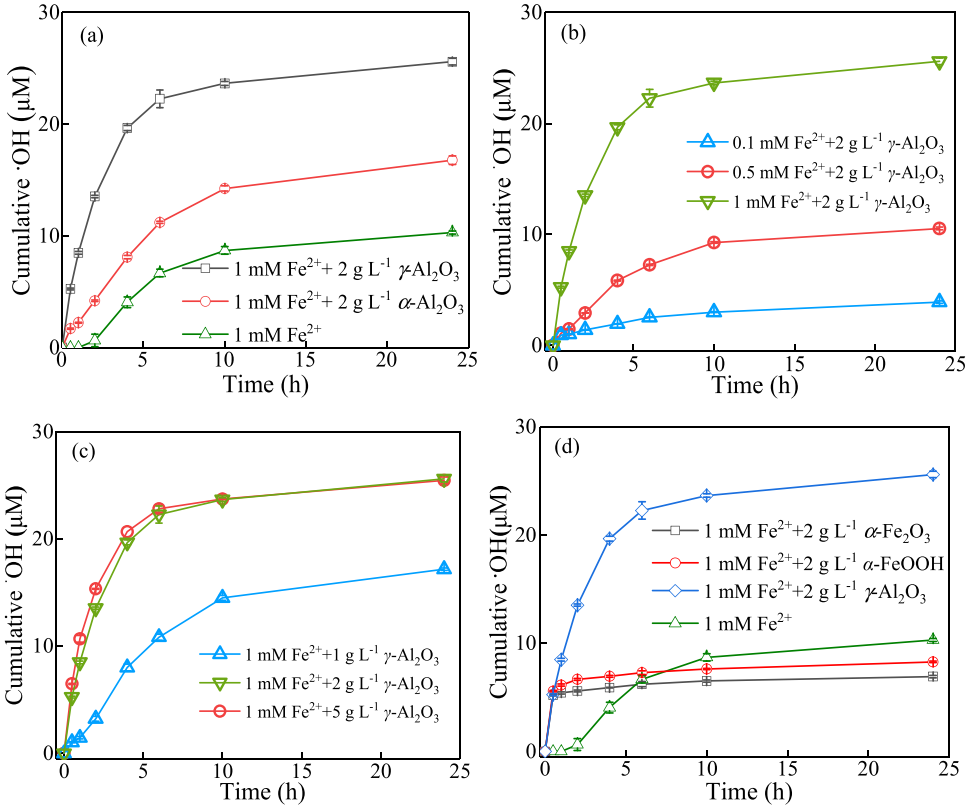


Fig. 1. a) and d) •OH accumulation during oxygenation of Fe(II) in the presence of different minerals; b) and c): the effects of Fe(II) concentration and γ-Al_2O_3 content on •OH accumulation. Reaction conditions: pH= 6 and 25 ± 1 °C. The Fe(II) concentrations shown in each figure were the initial concentration.

versus Fe^{2+}_{aq}only. The •OH production increased rapidly within 10 h and became slower afterwards (Fig. 1a), with 16.7 and 25.6 μM •OH produced in the α/γ-Al_2O_3 systems after 24 h, respectively. The rate of •OH production in α/γ-Al_2O_3 systems within the initial 2 h well followed the zero-order kinetics (R^2 >0.97), which showed that $k_{obs,\alpha\text{-}Al2O3}$= 1.99 ± 0.24 μM h^{-1} was much lower than that of $k_{obs,\gamma\text{-}Al2O3}$= 6.76 ± 0.44 μM h^{-1}. These results indicated that the presence of γ-Al_2O_3 significantly accelerated the •OH formation compared to α-Al_2O_3 during oxygenation.

With increasing Fe(II) concentration from 0.1 to 1 mM, the $k_{obs,\cdot OH}$ increased from 0.71 ± 0.22–6.76 ± 0.44 μM h^{-1}, with •OH production enhanced from 3.9 to 25.6 μM (Fig. 1b). These results indicated that increased Fe(II) concentrations led to the production of more •OH. With the augment of γ-Al_2O_3 dosages from 1 to 2 g L^{-1}, k_{obs} of •OH formation increased from 1.66 ± 0.23–6.76 ± 0.44 μM h^{-1} (Fig. 1c), and slightly increased to 7.66 μM h^{-1} at 5 g L^{-1}. However, •OH accumulation (25.4 μM at 5 g L^{-1}) was similar to that of 2 g L^{-1} (25.6 μM) within 24 h. These combined results indicated that •OH production significantly increased with the increasing γ-Al_2O_3 dosages from 1 to 2 g L^{-1} probably due to the enhanced $Fe(II)_{sorbed}$ oxidation, but changed negligibly with further addition of 5 g L^{-1} γ-Al_2O_3 perhaps owing to the limited Fe(II) contents.

Adding same dosages of α-Fe_2O_3/α-FeOOH, •OH accumulated rapidly within 1 h and then changed slightly, with 6.9 and 8.3 μM •OH produced after 24 h, respectively (Fig. 1d). The •OH accumulation in the presence of α-Fe_2O_3/α-FeOOH was much lower than that in the α/γ-Al_2O_3 systems (16.7 and 25.6 μM, respectively), probably suggesting that the Fe-bearing minerals inhibited Fe(II) oxidation and discussed in later section. The •OH formation was also confirmed with the electron paramagnetic resonance (EPR) using 5,5-dimethyl-1-pyrrolidine N-oxide (DMPO) as a spin-trapping agent. Results showed that DMPO-OH signal appeared during Fe(II) oxidation in the presence of α-Al_2O_3/α-FeOOH (Fig. S2c), indicating the production of •OH.

3.2. Changes of Fe species during oxygenation

To explore the process of •OH formation, changes of Fe species were examined during oxygenation. For Fe^{2+}_{aq} only, results showed that Fe^{2+}_{aq} changed slightly within 2 h and then decreased rapidly (Fig. S3), with 55.53 mg L^{-1} Fe(II) oxidized after 24 h. With the addition of α-Al_2O_3, Fe^{2+}_{aq} decreased quickly within the initial 4 h and slower afterwards (Fig. 2a). The Fe(II) oxidation within the initial 6 h well fitted the pseudo-first-order kinetics (R^2 >0.88), which showed that the $k_{obs,\alpha\text{-}Al2O3}$= 0.38 ± 0.03 h^{-1} was much higher than that of Fe^{2+}_{aq} only (k_{obs} =0.27 ± 0.04 h^{-1}). These results indicated that the presence of α-Al_2O_3 greatly enhanced the Fe^{2+}_{aq} oxidation. To further investigate the dynamic changes of different Fe species, chemical extraction was applied as depicted in Section 2.3. Results showed that $Fe(II)_{sorbed}$ or Fe(II) in lower-crystallinity quickly increased to 9.6 mg L^{-1} after 1 h and then declined gradually. The Fe(II) phases in the higher-crystallinity fraction also increased rapidly to 16.6 mg L^{-1} within the initial 4 h and then declined, which showed the retarded Fe(II) oxidation versus the $Fe(II)_{sorbed}$ or Fe(II) phases in lower-crystallinity (Fig. 2a). After 10 h oxidation, more than 94% of total Fe(II) was oxidized in the α-Al_2O_3 system, which was accompanied with the 85% of •OH accumulation (Fig. 1a). These results indicated that oxidation of Fe(II) species was responsible for •OH formation.

In γ-Al_2O_3 system, the k_{obs} of Fe^{2+}_{aq} (0.64 ± 0.07 h^{-1}) was much higher than that in α-Al_2O_3 system, suggesting that the enhanced Fe(II) oxidation occurred. These results corresponded well to the higher •OH accumulation in the γ-Al_2O_3 than in the α-Al_2O_3 system (Fig. 1a), indicating that the •OH formation was highly affected by Fe(II) oxidation due to the discrepant properties of Al_2O_3 minerals. The changes of $Fe(II)_{sorbed}$ or low-crystalline Fe(II) phases in the γ-Al_2O_3 system showed a

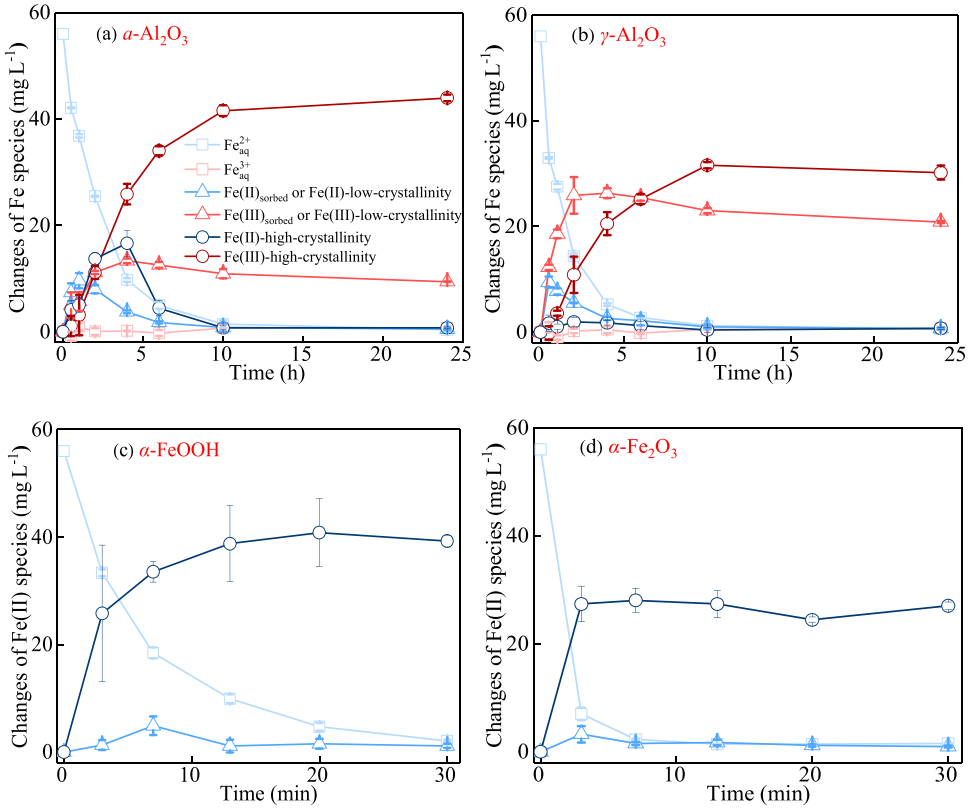


Fig. 2. Changes of different Fe species in a) α-Al_2O_3 and b) γ-Al_2O_3 systems, and Fe(II) species in c) α-FeOOH and b) α-Fe_2O_3 systems (pH 6.0). The surface Fe in lower-crystallinity was obtained by the subtraction of aqueous Fe from 0.5 M HCl-extracted Fe. Likewise, the surface Fe in higher-crystallinity was based on the subtraction of 0.5 M HCl-Fe from 5 M HCl-extracted Fe.

similar trend to that in the α-Al_2O_3 system, while limited $Fe(II)_{sorbed}$ or high-crystalline Fe(II) species (<1.9 mg L^{-1}) formed in the γ-Al_2O_3 system. After 24 h oxidation, more than 96% of total Fe(II) was oxidized, which was similar to that in α-Al_2O_3 system.

The changes of Fe(III) species were also monitored during oxygenation. Specifically, limited aqueous Fe(III) (<0.98 mg L^{-1}) was detected in the α/γ-Al_2O_3 systems (Figs. 2a and 2b). In α-Al_2O_3 system, the $Fe(III)_{sorbed}$ or low-crystalline Fe(III) phases increased to 13.3 mg L^{-1} within the initial 4 h and then gradually declined to 9.4 mg L^{-1} after 24 h oxidation. Correspondingly, the high-crystalline Fe(III) phases increased continuously to 41.5 mg L^{-1} within 10 h and then slightly changed, which was in accordance with the oxidation of total Fe(II) species. These combined results suggested that quick accumulation of low-crystalline Fe(III) phases occurred in the initial stage due to the rapid oxidation of Fe(II) and were then accompanied with the transformation into high-crystalline phases during oxygenation. Compared to the α-Al_2O_3 systems, similar trends of Fe(III) species were observed in the γ-Al_2O_3 systems. However, higher contents of $Fe(III)_{sorbed}$ or low-crystalline Fe(III) species (20.8 mg L^{-1}) were monitored in the γ-Al_2O_3 than in the α-Al_2O_3 system, which consequently induced less accumulation of high-crystalline Fe(III) species in the γ-Al_2O_3 system (Fig. 2b). These combined results indicated that the transformation of low-crystalline Fe(III) phases was strongly affected by the crystallinity of Al_2O_3 probably due to discrepant Fe(II) oxidation processes, which will be discussed later.

In the α-FeOOH/α-Fe_2O_3 systems, changes of Fe(II) rather than Fe (III) species were examined due to the interference of Fe(III) from the minerals themselves. Results showed that Fe^{2+}_{aq} decreased quickly (k_{obs}=0.15 ± 0.07 min^{-1}) in the α-FeOOH system. Concomitantly, $Fe(II)_{sorbed}$ or high-crystalline Fe(II) species slightly increased and then declined, while accumulation of high-crystalline Fe(II) phases was observed during oxygenation. Similar results were observed in the α-Fe_2O_3 system, but the k_{obs} of Fe^{2+}_{aq} (0.45 ± 0.11 min^{-1}) was much higher than that in the α-FeOOH system, which also showed the rapid formation of high-crystalline Fe(II) during oxygenation. After 24 h oxidation, 30.1% and 51.7% of total Fe(II) were oxidized in the α-FeOOH and the α-Fe_2O_3 systems much lower than that in the α/γ-Al_2O_3 systems, which was also well consistent with the lower $^{\bullet}OH$ production (Fig. 1c). These results suggested that $Fe(II)_{sorbed}$ can be preserved as high-crystalline Fe(II) phases on the α-FeOOH/α-Fe_2O_3 minerals that against oxidation.

3.3. Characterization of Fe species

To examine variations of near-surface Fe(II) species, XPS characterization was applied during oxygenation as XPS only detects the depth of serval nanometers (Yuan et al., 2018). After 2 h of oxidation, the fraction of surface-Fe(II) increased to 14.6% (Fig. S4a), indicating Fe(II) sorption on the γ-Al_2O_3 surface. Meanwhile, there was 85.4% of Fe(III) species formed, suggesting precipitation of Fe(III) occurred on the γ-Al_2O_3 surface. Thereafter, fraction of surface-Fe(II) quickly decreased to 2.8% after 10 h and disappeared after 24 h (Fig. S4b and S4c), which was roughly consistent with the changes of total Fe(II) species during oxygenation (Fig. 2b). Similar results were also observed in the α-Al_2O_3 system (Fig. S5). These results further proved that Fe(II) oxidation occurred on the Al_2O_3 surface. Compared to Al_2O_3, the fractions of surface-Fe(II) increased from 8.3% and 13.7–11.0% and 16.5% in the α-Fe_2O_3 and in the α-FeOOH systems, respectively (Figs. S6a-d), which corresponded to the accumulation of high-crystalline Fe(II) phases during oxygenation (Fig. 2c-d) and discussed in later section. Besides, the higher fraction of surface-Fe(II) in the α-FeOOH than the α-Fe_2O_3 system further indicated the higher retention of $Fe(II)_{sorbed}$ on α-FeOOH surface, probably due to its lower-crystallinity that induced quick recrystallization after Fe(II) sorption (Chen et al., 2020; Winkler et al., 2018).

To further identify the transformation of Fe phases, γ-Al_2O_3 particles were examined by Mössbauer spectroscopy. Results showed that 5.6% of QSD-Fe(II) (Fe(II) species in the quadrupole splitting distribution, Table S1) was present after 2 h oxidation and then disappeared after 24 h (Fig. 3), which was consistent with the changes of surface-Fe(II) as observed by XPS. These combined results suggested that an increasing fraction of Fe(II) was adsorbed on the γ-Al_2O_3 surface within the initial period and then completely oxidized by O_2. Meanwhile, there were 9.3% of QSD-Fe(III), 40.2% of HFD-Lp (lepidocrocite in the hyperfine field distribution), 32.0% of HFD-Gt (goethite) and 12.8% of the most disordered Fe(III) phases presented on γ-Al_2O_3 surface after 2 h oxidation (Fig. 3a), which indicated that the oxidized surface-Fe(II) transformed into these Fe(III) species, especially the high accumulation of HFD-Lp and HFD-Gt phases. After 24 h oxidation, fractions of HFD-Lp and the most disordered Fe(III) phases declined by 4.6% and 3.5%, respectively, which was accompanied with the increasing fractions of QSD-Fe(III) (1.3%) and HFD-Gt (12.5%) respectively (Fig. 3b). These results indicated that the low-crystalline Fe phases (e.g., HFD-Lp and the most disordered Fe phases) transformed into the high-crystalline Fe phases (e.g., HFD-Gt) during oxygenation, which was coincided with the increase of high-crystalline Fe phases based on the chemical dissolution (Fig. 2b). In terms of energy theory, the Gibbs free energies of thermodynamically stable Fe phases were more negative than metastable Fe phases (Huang et al., 2021b). Hence, transformation of low-crystalline Fe phases (e.g., ferrihydrite and lepidocrocite) to the high-crystalline Fe phases (e.g., goethite) was more thermodynamically favorable.

3.4. Mechanisms of $^{\bullet}OH$ production and Fe(II) oxidation

3.4.1. Relations between $^{\bullet}OH$ production and Fe(II) oxidation

Although more than 94% of the Fe(II) was oxidized within 24 h in the α/γ-Al_2O_3 systems, $^{\bullet}OH$ accumulation in the γ-Al_2O_3 system was much higher than in the α-Al_2O_3 system, indicating the different efficiency of $^{\bullet}OH$ production. Hence, correlation analyses between $^{\bullet}OH$ accumulation and the amount of oxidized Fe(II) were performed to explore the inherent discrepancies. Results showed that the oxidized Fe (II) was positively ($p < 0.01$) correlated with $^{\bullet}OH$ accumulation in these reaction systems (Fig. 4), proving that Fe(II) oxidation was responsible for $^{\bullet}OH$ production. Besides, the slope (representing the efficiency of $^{\bullet}OH$ production via Fe(II) oxidation) of Fe^{2+}_{aq} only (0.17 ± 0.02 μmol mg^{-1}) was much lower than that of the α-Al_2O_3 (0.26 ± 0.03 μmol mg^{-1}) and γ-Al_2O_3 (0.46 ± 0.03 μmol mg^{-1}) systems, indicating enhanced $^{\bullet}OH$ production efficiency in the presence of Al_2O_3 minerals. Two reasons might be responsible for the observed results. One is that the presence of Al_2O_3 minerals significantly enhanced Fe(II) oxidation and thus induced quick $^{\bullet}OH$ accumulation on the Al_2O_3 surface (Figs. 1 and 2), probably ascribed to the "surface confinements" (Fe(II) confined on the Al_2O_3 facets that can significantly promote $^{\bullet}OH$ production) compared to homogeneous reaction (without mineral or solids) in treatment of Fe^{2+}_{aq} only (Huang et al., 2016; Vikesland et al., 2007). Another possible reason is that the surface acid centers (e.g., ≡Al-OH, releasing H^+ via ionization) of Al_2O_3 particles catalyzed H_2O_2 formation via the enhanced protonation of superoxide anions (Tournassat et al., 2016), which consequently induced more $^{\bullet}OH$ production.

Besides, the $^{\bullet}OH$ production efficiency in the γ-Al_2O_3 system was much higher than that in the α-Al_2O_3 system, which was ascribed to the different properties of these minerals. Generally, the crystallinity of α-Al_2O_3 was higher than that of γ-Al_2O_3 at room temperature (Liu et al., 2008), indicating the lower-crystallinity of γ-Al_2O_3 compared to α-Al_2O_3. Additionally, the SSA of γ-Al_2O_3 (122.6 ± 0.6 m^2 g^{-1}) was much higher than that of α-Al_2O_3 (39.1 ± 0.9 m^2 g^{-1}), indicating that γ-Al_2O_3 provided more sites for Fe(II) sorption and consequently induced more $Fe(II)_{sorbed}$ available for oxidation and $^{\bullet}OH$ production. After Fe(II) sorption, Eh value of γ-Al_2O_3 system (−74.1 mV) was also more negative than that of α-Al_2O_3 system (−69.1 mV). Hence, more Fe^{2+}_{aq} was adsorbed on γ-Al_2O_3 than α-Al_2O_3 (Fig. S7a), which further caused the increased Fe(II) oxidation and $^{\bullet}OH$ accumulation probably

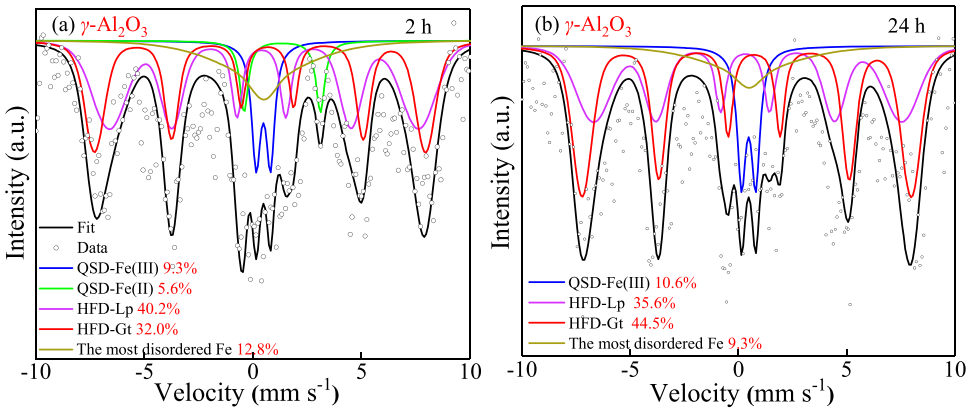


Fig. 3. Fitted Mössbauer spectra (13 K) of the Fe species during oxygenation of Fe(II) in the presence of γ-Al_2O_3. In each spectrum, the black line is the total calculated fit, through the discrete data points. The resolved spectral components and assignments are: (1) QSD-Fe^{III} (blue), Fe(III) phases not magnetically ordered (+ paramagnetic Fe^{III} (oxy-)hydroxides); (2) Fe^{II} (green), Fe^{II} in primary minerals or sorbed; (3) HFD-Lp (pink), Fe^{III} in lepidocrocite; (3) HFD-Gt (red), Fe^{III} in goethite; (4) The most disordered Fe pahses, Fe^{III} (oxyhydr)oxides near their blocking temperature (dark yellow, (T_N < collection temp)). The detailed fitting parameters are presented in Table S1.

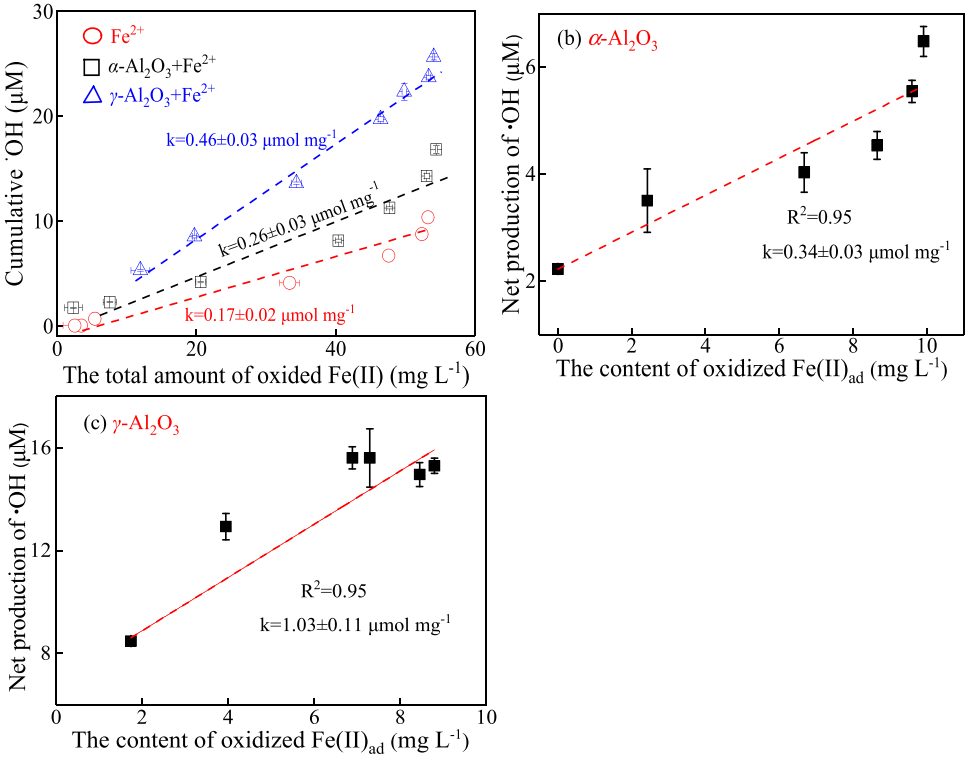


Fig. 4. Correlation analyses between the content of oxidized Fe(II) and •OH accumulation in the systems with/without Al_2O_3; b) and c) correlation analyses between the content of oxidized surface-Fe(II) ($Fe(II)_{ad}$) and net production of •OH (calculated from the subtraction of •OH production in CK from aqueous Fe(II) and α/γ-Al_2O_3 systems).

owing to the enhanced H_2O_2 formation and "surface confinements" induced by its lower-crystallinity. To further examine the role of surface-Fe(II) in driving •OH production, correlation analyses between the contents of oxidized surface-Fe(II) and net •OH production were performed. Results showed that contents of oxidized surface-Fe(II) were highly correlated (R^2 >0.95) with net •OH production (Figs. 4b and 4c), indicating the increased •OH production was mainly derived from oxidation of surface-Fe(II) species.

In the α-FeOOH and α-Fe_2O_3 systems, large amounts of Fe(II) was presented as the high-crystalline phases, which induced less Fe(II) oxidation and •OH production versus Al_2O_3 systems (Figs. 1 and 2). In contrast to Al_2O_3 systems, once Fe^{2+}_{aq} adsorbed to the surface of α-FeOOH and α-Fe_2O_3, rapid electron-transfer processes between $Fe(II)_{sorbed}$ and Fe(III) in lattice can occur and induce transformation and re-crystallization of these Fe oxides (Williams and Scherer, 2004; Yanina and Rosso, 2008; Zinder et al., 1986), which might concomitantly preserve Fe(II) species in high-crystalline minerals. The produced Fe(II) phases in higher-crystallinity were hardly to be identified based on spectroscopic measurements (e.g., Mössbauer analyses) due to their low contents and easy transformation during sample preparation. Besides, most studies have examined the Fe(II) species derived from Fe^{2+}_{aq} sorption onto Fe(III)-oxides (e.g., goethite and hematite) using Mössbauer spectra etc. (Larese-Casanova and Scherer, 2007; Latta et al., 2012; Notini et al., 2018, 2019; Rosso et al., 2010), which showed that Fe(II) species mainly existed as the doublets in spectrum and hardly to be specifically resolved. Even though, Larese-Casanova and Scherer (2007) have also proposed the identities of Fe(II) species formed during Fe^{2+}_{aq}sorption on the Hematite, which indicated that ≡Fe(III)OFe(II)OH or ≡Fe(III)OFe(II)$^+$ were the most likely candidates based on surface complexation modeling. Indeed, crystalline Fe-oxides (e.g., goethite and hematite) can adsorb large amounts Fe^{2+}_{aq} on its surface (Fig. S7), which further induced quick electron-transfer from $Fe(II)_{sorbed}$ to Fe(III)

in lattice, and caused the dissolution and reprecipitation of Fe-oxides to form thermodynamically more stable phases. Concurrently, $Fe(II)_{sorbed}$ can also be oxidized by O_2 to form low-crystalline phases (e.g., ferrihydrite) and then underwent the Fe(II)-catalyzed transformation into high-crystalline phases (e.g., goethite) (Hansel et al., 2005). These phases transformation further induced the surface coating and passivation on Fe-oxides and consequently inhibited continued electron injection into the bulk structure. The formed stable phase and surface passivation might preserve these Fe(II) species (e.g., ≡Fe(III)OFe(II)OH), which leads to the formation of stable Fe(II) phases (more studies need to examine its identities) (Larese-Casanova and Scherer, 2007). Hence, stable Fe(II) species (e.g., $(\equiv Fe^{III}OFe^{II})^{+}$) might be formed on the α-FeOOH/α-Fe_2O_3 surface, which consequently inhibited the Fe(II) oxidation and $^{\bullet}OH$ production.

3.4.2. The pathways for $^{\bullet}OH$ production

To examine the involvement of H_2O_2, CAT was added into reaction systems as scavenger (Zeng et al., 2021). Results showed that 100 mg L^{-1} CAT significantly decreased $^{\bullet}OH$ accumulation by 52.2% and 95.2% in the α-Fe_2O_3 and γ-Al_2O_3 systems, respectively (Fig. S8a), which proved that H_2O_2 was key intermediate ROS for $^{\bullet}OH$ production (Liao et al., 2019). To further examine the electron-transfer pathways from O_2 to H_2O_2, NBT, as the efficient scavenger for $O_2^{\bullet-}$ (Ryu et al., 2013), was added to investigate $^{\bullet}OH$ production. Results showed that negligible changes of $^{\bullet}OH$ accumulation was observed after 24 h in the α-Fe_2O_3 and γ-Al_2O_3 systems (Fig. S8b), indicating that two-electron transfer pathway was mainly responsible for the reduction of O_2 to H_2O_2 (Cheng et al., 2016). Several studies documented that $Fe(II)_{sorbed}$ rather than Fe^{2+}_{aq} was highly correlated with contaminants reduction (Amonette et al., 2000; Chun et al., 2005; Klupinski et al., 2004), in which the two-electron transfer mechanism on mineral surface was proposed to occur (Amonette et al., 2000; Wehrli, 1990). Hence, our results might also provide the evidence for two-electron transfer mechanism that occurred during oxygenation of $Fe(II)_{sorbed}$.

3.4.3. Fe mineral transformation

The presence of Al_2O_3 surface enhanced Fe(II) oxidation compared to Fe^{2+}_{aq} only (Fig. 2 and S3), which also favored the formation of low-crystalline Fe phases (Steefel and Van Cappellen, 1990), especially HFD-Fh and HFD-Lp (Fig. 3) (Carlson and Schwertmann, 1990). In our study, 40.2% of HFD-Lp and 12.8% of the most disordered Fe(III) phases formed on the γ-Al_2O_3 surface after 2 h, indicating the formation of low-crystalline Fe phases probably due to the "template effects" of γ-Al_2O_3 for *de novo* precipitates (Larese-Casanova et al., 2012; Latta et al., 2012). Besides, 32.0% of HFD-Gt was observed in the initial stage, indicating the rapid formation of high-crystalline Fe phases occurred on γ-Al_2O_3 surface. Previous studies documented that γ-Al_2O_3 was easily linked to the corner-sharing rows of Fe(III) hydroxide octahedral (the precursor to Gt phases). In this case, subsequent Fe(II) adsorption on the surface of precursors may further induce the growth of Gt instead of γ-Fe_2O_3 (Larese-Casanova et al., 2012). These mechanisms might be responsible for the observed formation of HFD-Gt in the initial stage.

Meanwhile, transformation of low-crystalline Fe phases (e.g., HFD-Lp) into high-crystalline Fe phases (e.g., HFD-Gt) was also monitored during oxygenation. The formation of Gt and Lp usually occurs via Ostwald ripening of ferrihydrite, with green rust (both soluble complexes and discrete precipitates) serving as a precursor to both Gt and Lp during oxidation of Fe^{2+}_{aq} (Cornell and Schwertmann, 2003). The $Fe(II)_{sorbed}$ was able to catalyze the rapid transformation from Lp to Gt phases, in which electron-transfer enhanced the dissolution and subsequent reprecipitation inducing the formation of more stable phases (Hansel et al., 2005, 2003; Yanina and Rosso, 2008). Hence, more accumulation of HFD-Gt than HFD-Lp phases were produced after oxygenation (Fig. 3b). However, more formation of high-crystalline Fe phases were observed in the α-Al_2O_3 than in the γ-Al_2O_3 systems (Fig. 2a-b). This was mainly because that Fe(II) oxidation rate highly affected the transformation of low-crystalline Fe phases during oxygenation at circumneutral conditions (Carlson and Schwertmann, 1990; Chen and Thompson, 2018). Due to the relatively slow oxidation of Fe(II) in the α-Al_2O_3 system relative to the γ-Al_2O_3 system (Fig. 2a), the longer exposure to surface-Fe(II) increased the recrystallization of low-crystalline Fe phases and then induced more formation of high-crystalline Fe phases (Chen and Thompson, 2018). Collectively, the Fe(II) oxidation, $^{\bullet}OH$ production and Fe phases transformation were highly influenced by the discrepant properties of Al_2O_3 minerals (e.g., SSA). Higher contents of high-crystalline Fe(II) phases were produced in the α-FeOOH/α-Fe_2O_3 systems than in the α/γ-Al_2O_3 systems, indicating more Fe(II) was preserved in high-crystalline phases. These discrepancies might be ascribed to the semiconductor properties of α-FeOOH/α-Fe_2O_3 (Becker et al., 2001; Hoffmann et al., 1995), which facilitated the electron-transfer reactions between surface-Fe(II) and Fe(III) in lattice compared to Al_2O_3 phases with electrical insulator properties (Barrón and Torrent, 2013). Hence, quick electron-transfer occurred once Fe^{2+}_{aq}sorbed on the surface of α-FeOOH/α-Fe_2O_3, which further induced the rapid recrystallization and preservation of sorbed Fe(II).

3.5. Effects of pH on $^{\bullet}OH$ formation

The effects of pH on $^{\bullet}OH$ formation were also investigated in the Fe(II)/γ-Al_2O_3 system, as pH strongly affected the Fe(II) speciation on mineral surface. The pH of interest for natural systems usually falls into the range of 5.0–7.5, thus $^{\bullet}OH$ production and Fe(II) oxidation were examined within this range. Results showed that $^{\bullet}OH$ accumulation increased continuously from 8.3 to 25.6 μM when increasing the pH from 5.0 to 6.0, and then decreased to 7.9 μM at pH 7.0 (Fig. 5a). These results indicated that $^{\bullet}OH$ production was highly affected by solution pH. Besides, $k_{obs,^{\bullet}OH}$ at pH 6.0 (8.5 μM h^{-1}, fitted with the initial three data points) was much higher than that at pH 5.0 (0.97 μM h^{-1}), suggesting the rate of $^{\bullet}OH$ formation increased with the increasing pH. However, $k_{obs,^{\bullet}OH}$ at pH 7.0 (6.3 μM h^{-1}) was much higher than that at pH 5.0, but the $^{\bullet}OH$ accumulation (7.9 μM) was similar to pH 5.0 (8.3 μM). These results implied that pH highly influenced the Fe(II) oxidation rate rather than its extent.

To further investigate the inherent discrepancies, changes of Fe species in the Fe(II)/γ-Al_2O_3 system were examined at different pH. At pH 5.0, limited Fe(II) (<0.50 mg g^{-1}) adsorbed on γ-Al_2O_3 probably due to the high zeta potential (Fig. S2b). Consequently, Fe^{2+}_{aq} decreased slightly due to the limited Fe(II) oxidation, which consequently induced negligible $^{\bullet}OH$ formation and Fe species transformation (Fig. 5 and S9). The changes of Fe species at pH 6.0 were described in Section 3.2, which showed that Fe^{2+}_{aq} decreased rapidly and induced the formation of Fe species in different crystallinities (Fig. 2b). Hence, the enhanced Fe(II) oxidation at pH 6.0 promoted more $^{\bullet}OH$ production compared to pH 5.0. At pH 7.0, $Fe(II)_{sorbed}$ significantly increased versus pH at 5.0 and 6.0 (Fig. S2b), which can be attributed to the Fe(II) adsorption by surface weak/strong-bonding sites based on the two-site models (Fig. 5b) (Li et al., 2009). Besides, surface of γ-Al_2O_3 was more negatively charged at pH 7.0 than that at pH 6.0 and 5.0, which facilitated the oxidation of surface Fe(II) due to the surface-complexation by active sites (e.g., ≡Al-O, ≡-$OH_{surface}$) (Huang et al., 2021b; Pignatello et al., 2006). Consequently, Fe(II) oxidation was too rapid to be examined at pH 7.0 (data not shown). Although most of Fe(II) (>98%) was oxidized at pH 7.0, $^{\bullet}OH$ production was to the low extent, which might be ascribed to the enhanced consumption of $^{\bullet}OH$ by active Fe(II) species. Hence, pH strongly affected Fe(II) speciation and its adsorption on mineral surfaces, which significantly impacted the Fe(II) oxidation and $^{\bullet}OH$ production.

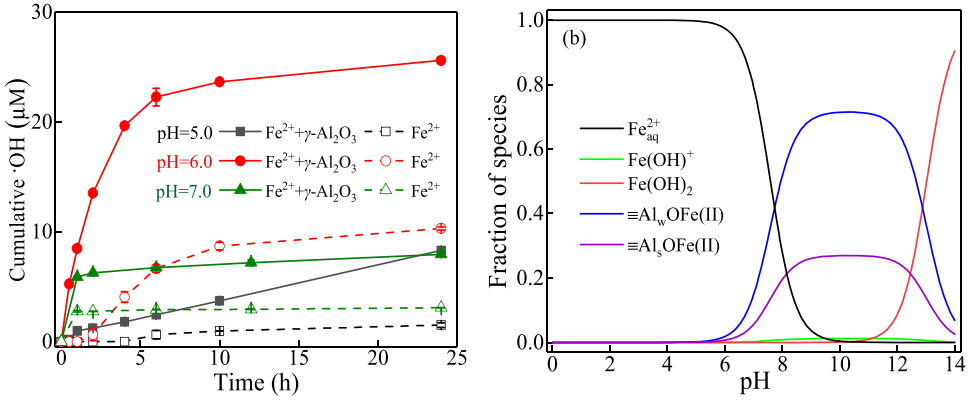


Fig. 5. Effects of different pH on the •OH production in Fe^{2+}/γ-Al_2O_3 systems; b) Fe(II) species in aqueous solution as a function of pH. ☰Al_wOFe(II): surface Fe(II) associated by weak-binding sites; ☰Al_sOFe(II): surface Fe(II) associated by strong-binding sites. Reaction conditions: $[minerals]_0 = 2\ g\ L^{-1}$, $[Fe^{2+}]_0 = 1$ mM at 25 ± 1 °C;.

3.6. Phenol degradation during oxygenation of Fe(II) in the presence of minerals

Negligible phenol degradation (<2.9%) was observed for Fe^{2+}_{aq}only (Fig. 6a), which might be attributed to the low •OH production and strong competition of Fe^{2+}_{aq} ($k_{Fe(II),\ \bullet OH} = 5 \times 10^8\ M^{-1}\ s^{-1}$) with phenol for •OH (Rose and Waite, 2002). However, phenol concentration rapidly decreased by 55.4% within 4 h in the Fe(II)/γ-Al_2O_3 system and then slightly declined, with 56.5% of phenol degraded after 24 h. With addition of 10% methanol as •OH scavenger, negligible phenol degradation was observed. With addition of the α-Al_2O_3 and α-Fe_2O_3, 37.0% and 32.0% of phenol were degraded after 24 h (Fig. 6b). The phenol degradation in the presence of different minerals was roughly consistent with •OH production (Fig. 1). These combined results proved that •OH was responsible for phenol degradation. With increasing Fe(II) concentrations from 0.1 to 1 mM, phenol degradation increased from 12.0% to 56.5%, but then decreased to 48.6% in the presence of 2 mM Fe(II) (Fig. 6c). The decrease of phenol degradation at high Fe(II) concentration can be ascribed to the intense competition from Fe^{2+}_{aq} with phenol for •OH (Chen et al., 2019). With increasing γ-Al_2O_3 dosages from 1 to 2 g L^{-1}, phenol degradation increased from 10.0% to 56.5%, but slightly changed (48.6%) in the presence of 5 g L^{-1} γ-Al_2O_3 (Fig. 6d). These results indicated that increased mineral dosages enhanced phenol degradation under the premise of sufficient $Fe_{2+\ aq}$.

4. Conclusions

Our study systematically explored the processes of Fe(II) oxidation

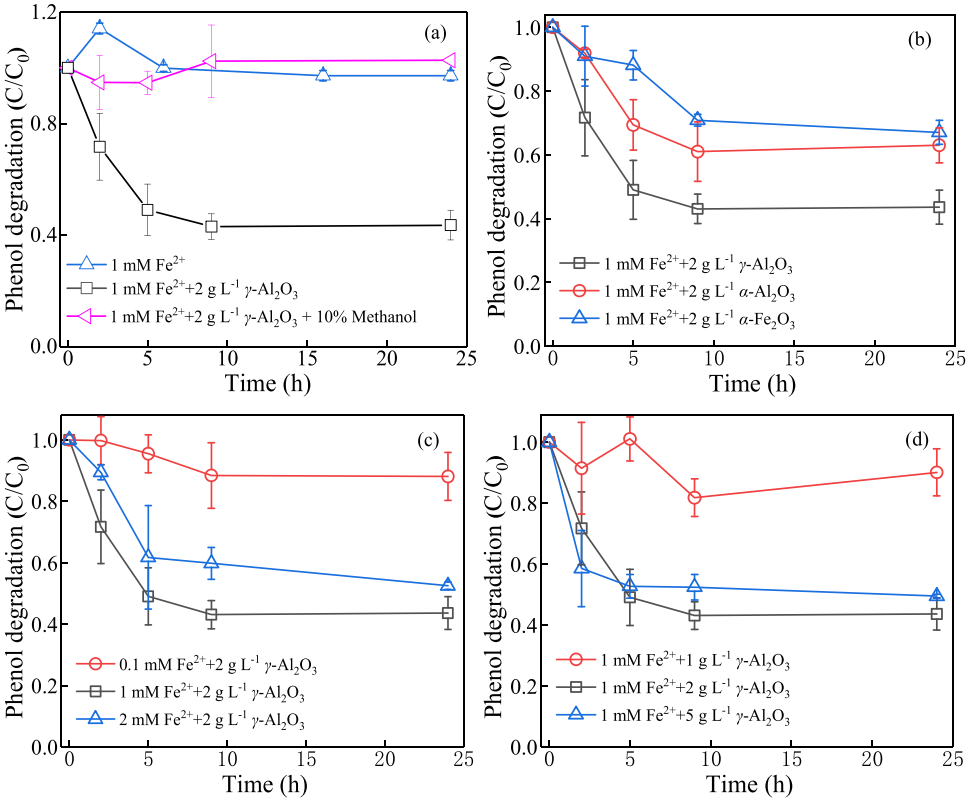


Fig. 6. a) Phenol degradation during oxygenation of Fe(II) under different conditions; b-d) Effects of mineral types, Fe(II) concentration and mineral contents on the phenol degradation. Reaction conditions: pH= 6 and 25 ± 1 °C.

and •OH production during oxygenation of $Fe(II)_{sorbed}$ on common minerals, which was significantly affected by the properties of minerals and solution conditions, such as SSA, electron-transfer capacity and pH. Enhanced •OH formation was observed during oxidation of Fe(II) in the presence of noniron oxides (i.e., α/γ-Al_2O_3) compared to iron oxides (i.e., α-FeOOH/α-Fe_2O_3) at circumneutral pH. The •OH accumulation was highly regulated by the surface-Fe(II) in α/γ-Al_2O_3 systems, especially $Fe(II)_{sorbed}$ or Fe(II) in low-crystalline minerals, which served as the main contributor to •OH production compared to other Fe(II) species. These findings might reveal the natural sources of •OH formation during surface-water/soil redox fluctuations and deepen the understanding of associated attenuation of organic pollutants. Because Fe(II)/Fe(III) serves as the important redox couple during geochemical processes, more studies need to further explore its environmental behaviors (e.g., ROS production) on minerals surface, as well as the effects of organic matters, which will be conducive to exploring the fate of nutrient elements (e.g., C and N) and contaminants in the critical zone of earth.

Environmental implications

This study systematically explored the processes of Fe(II) oxidation and •OH production during oxygenation of adsorbed Fe(II) on common minerals (i.e., α/γ-Al_2O_3, α-FeOOH/α-Fe_2O_3), which might reveal the natural sources of •OH formation during surface-water/soil redox fluctuations and deepen the understanding of associated attenuation of organic pollutants. Because Fe(II)/Fe(III) serves as the important redox couples during geochemical processes, more studies need to further explore its environmental behaviors (e.g., ROS production) on minerals surface, especially coupling with the organic matters, which will be conducive to exploring the fate of nutrient elements and contaminants in the critical zone of earth.

CRediT authorship contribution statement

Ning Chen: Preparation, Writing – review & editing. **Mengyuan Geng:** Data curation, Writing – review & editing. **Danyu Huang:** Writing – review & editing. **Mengxi Tan**: Data curation. **Zipeng Li**: Supply materials. **Guangxia Liu:** Writing – review & editing. **Changyin Zhu**: Conceptualization, Validation. **Guodong Fang:** Writing – review & editing. **Dongmei Zhou:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2022.128861.

References

Amonette, J.E., Workman, D.J., Kennedy, D.W., Fruchter, J.S., Gorby, Y.A., 2000. Dechlorination of carbon tetrachloride by Fe(II) associated with goethite. Environ. Sci. Technol. 34 (21), 4606–4613.

Barrón, V. and Torrent, J., 2013. Iron, manganese and aluminium oxides and oxyhydroxides, Mineralogical Society of Great Britain and Ireland.

Becker, U., Rosso, K.M., Hochella, M.F., 2001. The proximity effect on semiconducting mineral surfaces: a new aspect of mineral surface reactivity and surface complexation theory? Geochim. Et Cosmochim. Acta 65 (16), 2641–2649.

Carlson, L., Schwertmann, U., 1990. The effect of CO_2 and oxidation rate on the formation of goethite versus lepidocrocite from an Fe(II) system at pH 6 and 7. Clay Min. 25 (1), 65–71.

Catalano, J.G., Fenter, P., Park, C., Zhang, Z., Rosso, K.M., 2010. Structure and oxidation state of hematite surfaces reacted with aqueous Fe(II) at acidic and neutral pH. Geochim. Et Cosmochim. Acta 74 (5), 1498–1512.

Chen, C., Hall, S.J., Coward, E., Thompson, A., 2020. Iron-mediated organic matter decomposition in humid soils can counteract protection. Nat. Commun. 11 (1), 2255–2267.

Chen, C., Kukkadapu, R., Sparks, D.L., 2015. Influence of coprecipitated organic matter on Fe^{2+}(aq)-catalyzed transformation of ferrihydrite: implications for carbon dynamics. Environ. Sci. Technol. 49 (18), 10927–10936.

Chen, C., Thompson, A., 2018. Ferrous iron oxidation under varying pO_2 levels: the effect of Fe(III)/Al(III) oxide minerals and organic matter. Environ. Sci. Technol. 52 (2), 597–606.

Chen, N., Fang, G., Liu, G., Zhou, D., Gao, J., Gu, C., 2018. The effects of Fe-bearing smectite clays on •OH formation and diethyl phthalate degradation with polyphenols and H_2O_2. J. Hazard. Mater. 357 (5), 483–490.

Chen, N., Fang, G., Liu, G., Zhou, D., Gao, J., Gu, C., 2019. The degradation of diethyl phthalate by reduced smectite clays and dissolved oxygen. Chem. Eng. J. 355 (1), 247–254.

Chen, N., Fu, Q., Wu, T., Cui, P., Fang, G., Liu, C., Chen, C., Liu, G., Wang, W., Wang, D., Wang, P., Zhou, D., 2021a. Active iron phases regulate the abiotic transformation of organic carbon during redox fluctuation cycles of paddy soil. Environ. Sci. Technol. 55 (20), 14281–14293.

Chen, N., Huang, D., Liu, G., Chu, L., Fang, G., Zhu, C., Zhou, D., Gao, J., 2021b. Active iron species driven hydroxyl radicals formation in oxygenation of different paddy soils: implications to polycyclic aromatic hydrocarbons degradation. Water Res. 203, 117484.

Cheng, D., Yuan, S., Liao, P., Zhang, P., 2016. Oxidizing impact induced by mackinawite (FeS) nanoparticles at oxic conditions due to production of hydroxyl radicals. Environ. Sci. Technol. 50 (21), 11646–11653.

Chun, C.L., Hozalski, R.M., Arnold, W.A., 2005. Degradation of drinking water disinfection byproducts by synthetic goethite and magnetite. Environ. Sci. Technol. 39 (21), 8525–8532.

Cornell, R.M., Schwertmann, U., 2003. The Iron Oxides: Structure, Properties, Reactions, Occurrences and Uses. John Wiley & Sons.

Dzombak, D.A., Morel, F., 1990. Surface Complexation Modeling: Hydrous Ferric Oxide. Wiley-Interscience, New York.

Frierdich, A.J., Helgeson, M., Liu, C., Wang, C., Rosso, K.M., Scherer, M.M., 2015. Iron atom exchange between hematite and aqueous Fe(II). Environ. Sci. Technol.

Gligorovski, S., Strekowski, R., Barbati, S., Vione, D., 2015. Environmental implications of hydroxyl radicals (•OH). Chem. Rev. 115 (24), 13051–13092.

Gorski, C.A., Edwards, R., Sander, M., Hofstetter, T.B., Stewart, S.M., 2016. Thermodynamic characterization of iron oxide–aqueous Fe^{2+} redox couples. Environ. Sci. Technol. 50 (16), 8538–8547.

Hansel, C.M., Benner, S.G., Fendorf, S., 2005. Competing Fe(II)-induced mineralization pathways of ferrihydrite. Environ. Sci. Technol. 39 (18), 7147–7153.

Hansel, C.M., Benner, S.G., Neiss, J., Dohnalkova, A., Kukkadapu, R.K., Fendorf, S., 2003. Secondary mineralization pathways induced by dissimilatory iron reduction of ferrihydrite under advective flow. Geochim. Et Cosmochim. Acta 67 (16), 2977–2992.

Heron, G., Crouzet, C., Bourg, A.C.M., Christensen, T.H., 1994. Speciation of Fe(II) and Fe(III) in contaminated aquifer sediments using chemical extraction techniques. Environ. Sci. Technol. 28 (9), 1698–1705.

Hiemstra, T., De Wit, J.C.M., Van Riemsdijk, W.H., 1989. Multisite proton adsorption modeling at the solid/solution interface of (hydr)oxides: a new approach: II. Application to various important (hydr)oxides. J. Colloid Interface Sci. 133 (1), 105–117.

Hoffmann, M.R., Martin, S.T., Choi, W., Bahnemann, D.W., 1995. Environmental applications of semiconductor photocatalysis. Chem. Rev. 95 (1), 69–96.

Huang, H., Ji, X., Cheng, L., Zhao, F., Wang, P., 2021a. Free radicals produced from the oxidation of ferrous sulfides promote the remobilization of cadmium in paddy soils during drainage. Environ. Sci. Technol. 55 (14), 9845–9853.

Huang, J., Jones, A., Waite, T.D., Chen, Y., Huang, X., Rosso, K.M., Kappler, A., Mansor, M., Tratnyek, P.G., Zhang, H., 2021b. Fe(II) redox chemistry in the environment. Chem. Rev. 13, 8161–8233.

Huang, J., Zhang, H., 2020. Redox reactions of iron and manganese oxides in complex systems. Front. Front. Environ. Sci. Eng. 14 (5), 76.

Huang, X., Hou, X., Zhao, J., Zhang, L., 2016. Hematite facet confined ferrous ions as high efficient Fenton catalysts to degrade organic contaminants by lowering H_2O_2 decomposition energetic span. Appl. Catal. B: Environ. 181, 127–137.

Jones, A.M., Griffin, P.J., Collins, R.N., Waite, T.D., 2014. Ferrous iron oxidation under acidic conditions -the effect of ferric oxide surfaces. Geochim. Et Cosmochim. Acta 145, 1–12.

Kandegedara, A., Rorabacher, D.B., 1999. Noncomplexing tertiary amines as "better" buffers covering the range of pH 3−11. Temperature dependence of their acid dissociation constants. Anal. Chem. 71 (15), 3140–3144.

Klupinski, T.P., Chin, Y.-P., Traina, S.J., 2004. Abiotic degradation of pentachloronitrobenzene by Fe(II): reactions on goethite and iron oxide nanoparticles. Environ. Sci. Technol. 38 (16), 4353–4360.

Larese-Casanova, P., Kappler, A., Haderlein, S.B., 2012. Heterogeneous oxidation of Fe (II) on iron oxides in aqueous systems: identification and controls of Fe(III) product formation. Geochim. Cosmochim. Acta 91, 171–186.
Larese-Casanova, P., Scherer, M.M., 2007. Fe(II) sorption on hematite: new insights based on spectroscopic measurements. Environ. Sci. Technol. 41 (2), 471–477.
Latta, D.E., Bachman, J.E., Scherer, M.M., 2012. Fe electron transfer and atom exchange in goethite: Influence of Al-substitution and anion sorption. Environ. Sci. Technol. 46 (19), 10614–10623.
Li, F.-b, Tao, L., Feng, C., Li, X., Sun, K., 2009. Electrochemical evidences for promoted interfacial reactions: the role of Fe(II) adsorbed onto γ-Al_2O_3 and TiO_2 in reductive transformation of 2-nitrophenol. Environ. Sci. Technol. 43 (10), 3656–3661.
Liao, P., Yu, K., Lu, Y., Wang, P., Liang, Y., Shi, Z., 2019. Extensive dark production of hydroxyl radicals from oxygenation of polluted river sediments. Chem. Eng. J. 368, 700–709.
Liu, X., Yuan, S., Tong, M., Liu, D., 2017. Oxidation of trichloroethylene by the hydroxyl radicals produced from oxygenation of reduced nontronite. Water Res. 113, 72–79.
Liu, Y., He, H., Mu, Y., 2008. . Heterogeneous reactivity of carbonyl sulfide on α-Al_2O_3 and γ-Al_2O_3. Atmos. Environ. 42 (5), 960–969.
Mopper, K., Zhou, X.L., 1990. Hydroxyl radical photoproduction in the sea and its potential impact on marine processes. Science 250 (4981), 661–664.
Nano, G.V., Strathmann, T.J., 2006. Ferrous iron sorption by hydrous metal oxides. J. Colloid Interface Sci. 297 (2), 443–454.
Notini, L., Latta, D.E., Neumann, A., Pearce, C.I., Sassi, M., N'Diaye, A.T., Rosso, K.M., Scherer, M.M., 2018. The role of defects in Fe(II)–goethite electron transfer. Environ. Sci. Technol. 52 (5), 2751–2759.
Notini, L., Latta, D.E., Neumann, A., Pearce, C.I., Sassi, M.J., N'Diaye, A.T., Rosso, K.M. and Scherer, M.M., 2019. A closer look at Fe(II) passivation of goethite. ACS Earth and Space Chemistry.
Pham, A.N., Waite, T.D., 2008a. Modeling the kinetics of Fe(II) oxidation in the presence of citrate and salicylate in aqueous solutions at pH 6.0−8.0 and 25 °C. J. Phys. Chem. A 112 (24), 5395–5405.
Pham, A.N., Waite, T.D., 2008b. Oxygenation of Fe(II) in natural waters revisited: Kinetic modeling approaches, rate constant estimation and the importance of various reaction pathways. Geochim. Et Cosmochim. Acta 72 (15), 3616–3630.
Pignatello, J.J., Oliveros, E., MacKay, A., 2006. Advanced oxidation processes for organic contaminant destruction based on the fenton reaction and related chemistry. Crit. Rev. Environ. Sci. Technol. 36 (1), 1–84.
Qian, A., Yuan, S., Xie, S., Tong, M., Zhang, P., Zheng, Y., 2019. Oxidizing capacity of iron electrocoagulation systems for refractory organic contaminants transformation. Environ. Sci. Technol. 53 (21), 12629–12638.
Rose, A.L., Waite, T.D., 2002. Kinetic model for Fe(II) oxidation in seawater in the absence and presence of natural organic matter. Environ. Sci. Technol. 36 (3), 433–444.
Rosso, K.M., Morgan, J.J., 2002. Outer-sphere electron transfer kinetics of metal ion oxidation by molecular oxygen. Geochim. Et Cosmochim. Acta 66 (24), 4223–4233.
Rosso, K.M., Yanina, S.V., Gorski, C.A., Larese-Casanova, P., Scherer, M.M., 2010. Connecting observations of hematite (α-Fe2O3) growth catalyzed by Fe(II). Environ. Sci. Technol. 44 (1), 61–67.
Ryu, J., Monllor-Satoca, D., Kim, D.H., Yeo, J., Choi, W., 2013. Photooxidation of arsenite under 254 nm irradiation with a quantum yield higher than unity. Environ. Sci. Technol. 47 (16), 9381–9387.
Shaw, T.J., Luther, G.W., Rosas, R., Oldham, V.E., Coffey, N.R., Ferry, J.L., Dias, D.M.C., Yücel, M. and Thibault de Chanvalon, A., 2021. Fe-catalyzed sulfide oxidation in hydrothermal plumes is a source of reactive oxygen species to the ocean. Proceedings of the National Academy of Sciences, USA 118(40), e2026654118.
Steefel, C.I., Cappellen, P, Van, 1990. A new kinetic approach to modeling water-rock interaction: the role of nucleation, precursors, and Ostwald ripening. Geochim. Cosmochim. Acta 54 (10), 2657–2677.
Tamura, H., Goto, K., Yotsuyanagi, T., Nagayama, M., 1974. Spectrophotometric determination of iron (II) with 1, 10-phenanthroline in the presence of large amounts of iron(III). Talanta 21, 314–318.
Tournassat, C., Davis, J.A., Chiaberge, C., Grangeon, S., Bourg, I.C., 2016. Modeling the acid-base properties of montmorillonite edge surfaces. Environ. Sci. Technol. 50 (24), 13436–13445.
Vikesland, P.J., Heathcock, A.M., Rebodos, R.L., Makus, K.E., 2007. Particle size and aggregation effects on magnetite reactivity toward carbon tetrachloride. Environ. Sci. Technol. 41 (15), 5277–5283.
Voelz, J.L., Johnson, N.W., Chun, C.L., Arnold, W.A., Penn, R.L., 2019. Quantitative dissolution of environmentally accessible iron residing in iron-rich minerals: a review. ACS Earth Space Chem. 3 (8), 1371–1392.
Wehrli, B., 1990. Aquatic Chemical Kinetics: Reaction Rates of Processes in Natural Waters. Wiley-Interscience, New York.
Williams, A.G.B., Scherer, M.M., 2004. Spectroscopic evidence for Fe(II)-Fe(III) electron transfer at the iron oxide-water interface. Environ. Sci. Technol. 38 (18), 4782–4790.
Winkler, P., Kaiser, K., Thompson, A., Kalbitz, K., Fiedler, S., Jahn, R., 2018. Contrasting evolution of iron phase composition in soils exposed to redox fluctuations. Geochim. Et Cosmochim. Acta 235, 89–102.
Wu, J., Zhao, X., Li, Z., Gu, X., 2020. Thermodynamic and kinetic coupling model of Cd (II) and Pb(II) adsorption and desorption on goethite. Sci. Total Environ. 727, 138730.
Xie, W., Yuan, S., Tong, M., Ma, S., Liao, W., Zhang, N., Chen, C., 2020. Contaminant degradation by·OH during sediment oxygenation: dependence on Fe(II) species. Environ. Sci. Technol. 54 (5), 2975–2984.
Yanina, S.V., Rosso, K.M., 2008. Linked reactivity at mineral-water interfaces through bulk crystal conduction. Science 320 (5873), 218–222.
Yuan, S., Liu, X., Liao, W., Zhang, P., Wang, X., Tong, M., 2018. Mechanisms of electron transfer from structrual Fe(II) in reduced nontronite to oxygen for production of hydroxyl radicals. Geochim. Et Cosmochim. Acta 223, 422–436.
Zeng, Q., Dong, H., Wang, X., Yu, T., Cui, W., 2017. Degradation of 1, 4-dioxane by hydroxyl radicals produced from clay minerals. J. Hazard. Mater. 331, 88–98.
Zeng, Q., Wang, X., Liu, X., Huang, L., Hu, J., Chu, R.K., Tolić, N., Dong, H., 2020. Mutual interactions between reduced Fe-bearing clay minerals and humic acids under dark, oxygenated condition: hydroxyl radical generation and humic acid transformation. Environ. Sci. Technol. 54, 15013–15023.
Zeng, Y., Fang, G., Fu, Q., Dionysiou, D.D., Wang, X., Gao, J., Zhou, D., Wang, Y., 2021. Photochemical characterization of paddy water during rice cultivation: formation of reactive intermediates for As(III) oxidation. Water Res. 206, 117721.
Zhang, P., Yuan, S., Chen, R., Bu, X., Tong, M., Huang, Q., 2020. Oxygenation of acid sulfate soils stimulates CO_2 emission: Roles of acidic dissolution and hydroxyl radical oxidation. Chem. Geol. 533, 119437–119449.
Zhao, Y., Xiang, W., Huang, C., Liu, Y., Tan, Y., 2021. Production of hydroxyl radicals following water-level drawdown in peatlands: a new induction mechanism for enhancing laccase activity in carbon cycling. Soil Biol. Biochem. 156, 108241.
Zhou, X., Mopper, K., 1990. Determination of photochemically produced hydroxyl radicals in seawater and freshwater. Mar. Chem. 30 (90), 71–88.
Zinder, B., Furrer, G., Stumm, W., 1986. The coordination chemistry of weathering: II. Dissolution of Fe(III) oxides. Geochim. Et Cosmochim. Acta 50 (9), 1861–1869.